

12th Ablation Workshop
Lexington, KY, USA, November 9-10, 2022

Identification and Study of Validation Level Test Cases for Computational Modeling of Non-Charring Ablators

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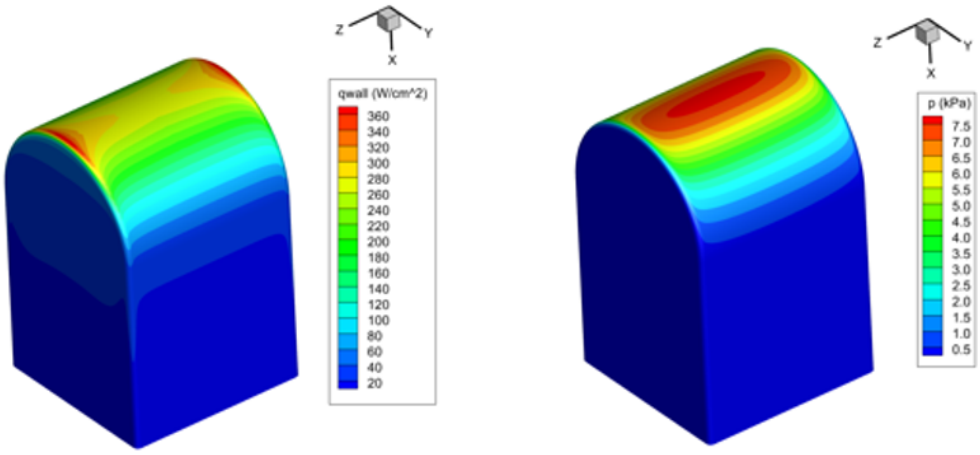
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Computational modeling of Thermal Protection System (TPS) materials, used for aerospace applications, provides numerous advantages in preliminary selection and design of a heatshield material and shape for atmospheric entry vehicles. However, to serve as a reliable tool for prediction of material thermal and ablative behavior, the modeling approach needs to be validated against real experimental and flight data, preferably at a range of applied conditions. The validation study is typically very complex as it requires reliable measured data not only for the material thermal response and surface recession, but also well characterized environmental conditions. The validation problem becomes even more complex when the material thermal response is dictated by multi-physics effects such as solid conduction, in-depth thermal decomposition, pyrolysis gas flow and chemical reactions. The multi-physics effects complicate not only the modeling effort, but also the experimental measurement for validation of various aspects of the highly coupled problem.

In this study, an attempt is made to identify suitable experimental data that could serve as a source for validation of material thermal response modeling tools. To reduce the computational complexity, this study focuses only on non-charring ablators, where the material thermal response could be modeled with a single governing equation for solid conduction and the ablation is limited only to the surface of the material. With a well characterized and publicly available experimental data being sparse, the study is limited in presenting test cases for only three materials: camphor, graphite and FiberForm® in the sequence of increased modeling complexity. Graphite is a commonly used TPS material for aerospace applications, both for leading edges of high-speed vehicles and internal insulation of solid rocket motors. FiberForm® is a porous carbon pre-form used in preparation of the well known PICA material Tran et al. [1996]. Inclusion of camphor into the list is conditioned with the relative simplicity in modeling the material thermal and chemical response and the low-enthalpy flow environment. In addition, camphor has been used as a simple test material for study of flow transition behavior by Stock and Ginoux [1973] and assessment of a heatshield shape change at flight relevant conditions by Rotondi et al. [2022].

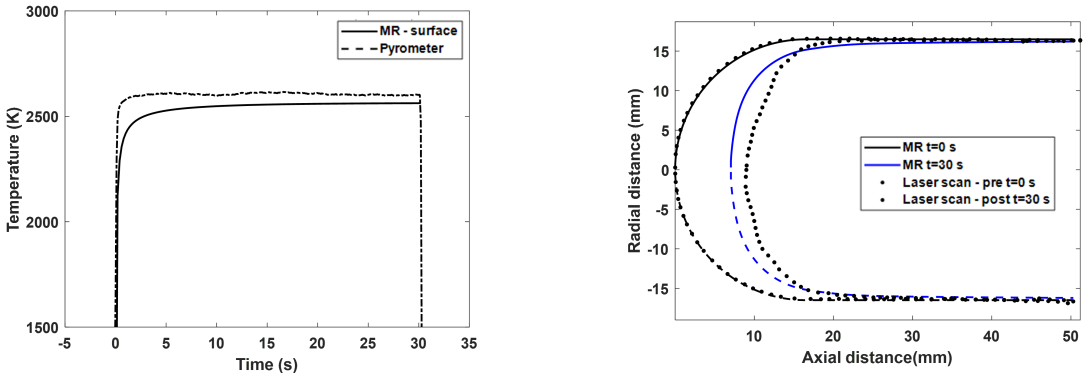
In this work, the identified experimental data was extracted from the public literature and test cases that yet have been published. As it was found from the review, not a single test case contains an exhaustive set of data that would validate every aspect of the material physics. However, in the data collected, various aspects of the material behavior can be still validated, such as surface and in-depth temperature, amount of recession and a shape change. The identified experimental data for each case is accompanied with a characterized flow environment and simulated boundary conditions predicted by a Data-Parallel Line Relaxation (DPLR) code Wright et al. [1998]. In addition, material thermal response numerical simulations in each test case are performed with Kentucky Aerothermodynamics and Thermal Response System (KATS-MR) Zibitsker et al. [2022] providing a comparative study and a sanity check for the proposed validation data.

Sample results from the performed numerical study are shown below. Figure 1 shows distribution of surface heat flux and pressure values on a hemi-cylinder model made of FiberForm® and tested in HyMETS arc-jet facility. The results are shown for the high pressure condition among the two tests. Flow simulation was performed with DPLR code on a quarter of original geometry. In the figure, the quarter shape was mirrored across zx and xy planes to show the complete distribution. Figure 2 shows the material response results for the high pressure case (7500 Pa), simulated with KATS-MR and a comparison to the experimental data for the surface temperature and shape shape. The simulation was performed on a 2-D slice, extracted in the xy plane at the middle of the sample. Figure 3 shows the material response simulation for the low pressure case (3500 Pa) and a comparison to the experimental data for surface temperature and shape change.



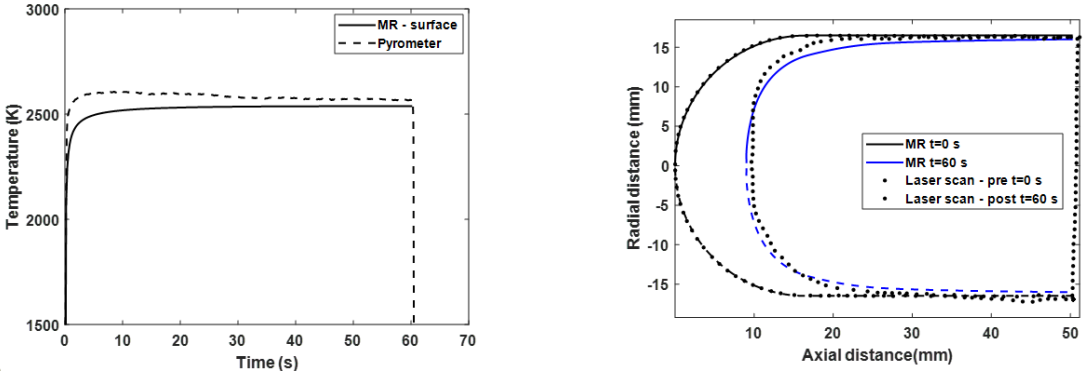
(a) Heat flux. (b) Pressure.

Figure 1: High pressure case. Simulated flow boundary conditions on hemi-cylinder model. Quarter shape was mirrored across zx and xy planes.



(a) Surface temperature. (b) Shape change.

Figure 2: FiberForm® - high pressure case. Comparison of simulated material response with experimental data.



(a) Surface temperature. (b) Shape change.

Figure 3: FiberForm® - low pressure case. Comparison of simulated material response with experimental data.

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